Extracting the Born cross section for $ep \to e'p'J/\psi$ at CLAS12: procedure, master formula, and the traps in our own samples

Campaign `prod_jpsi_f18in_20261009`

`exclurad_py` note, branch `jpsi-born-xsec-note`

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Abstract

What to do with a data yield binned in (W, Q^2, t), a radiative Monte Carlo and a Born Monte Carlo in order to obtain a Born cross section. The master formula is given in two algebraically equivalent forms. The one to *compute* is the ratio of the data yield to the reconstructed radiative Monte Carlo, times the model: in that form a wrong model *normalisation* cancels exactly, which matters here because `jpsi.diffrad_vm` sits a factor 4–6 below the data. Only the shape inside the bin survives, and it is iterated away. Six inputs beyond the three samples are required and are listed explicitly; the branching ratio $\mathcal{B}(J/\psi \to e^+e^-)$ is one of them and is *not* applied anywhere in the generator, a factor 16.7 if missed. Five properties of the existing campaign were verified while writing this note and are reported in Sec. 5, including the fact that the radiative and Born cards differ in two fields rather than one, that the Born outbending arm is 28 % short of the radiative arm, and that the generator's `sigma_nb` cannot be used to form the radiative correction.

1 The inputs

Three samples are not enough. The complete set is Table 1. Items 4–10 are not refinements: each is a factor, or a bias of the order of the effect being measured.

#	Input	Source
1	data yield in (W, Q^2, t) bins	analysis ntuple
2	radiative MC through GEMC	`prod_jpsi_f18in_20261009/rad`
3	Born MC through GEMC	`prod_jpsi_f18in_20261009/born`
4	integrated luminosity $\mathcal{L}$	for exactly the run/trigger selection that made the yield
5	signal extraction prescription	$M(ee)$ window and background shape
6	$\mathcal{B}(J/\psi \to e^+e^-) = (5.971 \pm 0.032)\,\%$	PDG. **Not in the generator**, Sec. 5.1
7	data/MC efficiency scale factors	FT electron, lepton and proton tracking, trigger
8	bin volumes, centroids, averaging rule	chosen once and written down
9	one cut list for data and *both* MC arms	exclusivity cut included
10	a model to iterate with	Sec. 3, step 7

Table 1: Inputs to the extraction.

2 The master formula

Each arm carries a per-event weight built from what the generator already reports in `<job>.config.json`,

$$w^{\text{arm}} = \frac{\sigma^{\text{arm}}_{\text{gen}}}{N^{\text{arm}}_{\text{gen}}} \quad [\text{nb/event}], \qquad \sigma_{\text{gen}} \equiv \texttt{report.sigma_nb}. \tag{1}$$

Let n_i be the number of *reconstructed* Monte Carlo events of a given arm falling in bin i after the identical analysis chain, and m_i the number of *generated* (truth) events in bin i with no cuts at all.

Form A — ratio to the radiative MC. This is the one to compute.

$$\boxed{\sigma_i^{\mathrm{Born}} = R_i\,\bar{\sigma}_i^{\mathrm{Born,model}}, \qquad R_i = \frac{N_i^{\mathrm{data}}}{\mathcal{L}\,\mathcal{B}\,w^{\mathrm{rad}}\,n_i^{\mathrm{rad}}}} \tag{2}$$

Form B — explicit factors. This is the one to tabulate systematics on.

$$\bar{\sigma}_i^{\mathrm{Born}} = \frac{N_i^{\mathrm{data}}}{\mathcal{L}\,\mathcal{B}\,\Delta\Omega_i\,\varepsilon_i^{\mathrm{Born}}\,C_i^{\mathrm{rad}}}, \qquad \varepsilon_i^{\mathrm{Born}} = \frac{n_i^{\mathrm{born}}}{m_i^{\mathrm{born}}}, \qquad C_i^{\mathrm{rad}} = \frac{w^{\mathrm{rad}}\,n_i^{\mathrm{rad}}}{w^{\mathrm{born}}\,n_i^{\mathrm{born}}}. \tag{3}$$

Equations (2) and (3) are the same once the bin-averaging is done consistently. Form A is what should be computed, for three reasons.

1. Acceptance, efficiency, resolution smearing, bin migration and radiation all sit inside one measured number, n_i^{rad}, obtained by running the same analysis code over reconstructed Monte Carlo. Nothing is factorised that does not factorise.

2. The $M(ee)$ window efficiency cancels. The same window is applied to the data and to n_i^{rad}, so it never has to be computed, and the radiative lineshape is simulated rather than assumed.

3. **A wrong model normalisation cancels exactly.** If the model is scaled by a factor f, then $n_i^{\mathrm{rad}} \to f\,n_i^{\mathrm{rad}}$, hence $R_i \to R_i/f$ and $\bar{\sigma}_i^{\mathrm{Born,model}} \to f\,\bar{\sigma}_i^{\mathrm{Born,model}}$, so

$$\sigma_i^{\mathrm{Born}} = R_i\,\bar{\sigma}_i^{\mathrm{Born,model}} \quad \text{is invariant under} \quad \sigma^{\mathrm{model}} \to f\,\sigma^{\mathrm{model}}. \tag{4}$$

The third point is not academic. The model `jpsi.diffrad_vm` carries $\sigma_T \propto Q^{-3.11}$, which double-counts the $1/Q^2$ already present in the virtual-photon flux, and it lies a factor 4–6 below the data at Forward Tagger kinematics. Form A is immune to that. What survives is only the *shape inside the bin*, and step 7 iterates it away.

3 Procedure

Step 1 — one cut list. Write the cuts once and apply them to the data, the radiative MC and the Born MC from the same code. Include the exclusivity cut. Both cards carry `vcut = 0`, so nothing was cut at generation and the whole radiative phase space is in the files; the exclusivity cut is applied *offline*, and that is the only way migration through resolution is simulated.

Step 2 — signal yield per bin. Count in a window around $M(ee) = 3.0969$ GeV. Do *not* fit a free lineshape: a free Gaussian width absorbs the radiative tail and biases the yield. Window counting reduces that model bias by about a factor 5. Subtract background from sidebands or a fitted continuum, and propagate the subtraction error into N_i^{data}.

Step 3 — the same window on the MC. The identical $M(ee)$ window applied to the reconstructed radiative MC gives n_i^{rad}. Both arms run `idecay = 2`, so the decay final-state radiation is in the simulation.

Step 4 — normalise. $Y_i^{\mathrm{rad}} = \mathcal{L}\,\mathcal{B}\,w^{\mathrm{rad}}\,n_i^{\mathrm{rad}}$. Average `sigma_nb` over jobs; all production jobs share the card, so w^{rad} is common to them.

Step 5 — form R_i and multiply by the model. Bin-average the model's Born cross section over the bin using the model's own density, *or* quote at the bin centroid with a bin-centring correction. State which. The two differ wherever the t-slope is steep, which near threshold is everywhere.

Step 6 — compute $C_i^{\rm rad}$ in any case. Form A does not need it, but it must be reported, and its bin-to-bin behaviour is the diagnostic that says whether the correction is sane. On this Monte Carlo we measure 0.80/0.76/0.62 by W bin, falling as W rises. A $C^{\rm rad}$ that is flat in W, or above unity, means something upstream is wrong.

Step 7 — iterate. Reweight the Monte Carlo event by event by $\sigma^{\rm extracted}/\sigma^{\rm model}$ as a function of (W, Q^2, t), then recompute $n_i^{\rm rad}$ and R_i. Two or three passes converge. This is required rather than cosmetic: the model is fitted over $Q^2 = 0.015$–0.1 GeV^2 while the generation box runs 0.001–0.25, so without iteration the bin-averaging is model-dominated at both Q^2 edges.

Step 8 — efficiency scale factors. GEMC does not reproduce the Forward Tagger electron or the tracking efficiencies. Apply the measured data/MC ratios to $Y_i^{\rm rad}$, per bin wherever they are kinematics-dependent.

Step 9 — quote it. For $ep \to e'p'J/\psi$ no flux factor is needed. If converting to $\sigma(\gamma^* p \to J/\psi\, p)$, state the flux convention (Hand or Gilman) explicitly: it moves the number.

4 Statistical error

From Form A, with the three counts independent,

$$\left(\frac{\delta R_i}{R_i}\right)^2 = \frac{1}{N_i^{\rm data}} + \frac{1}{n_i^{\rm rad}} + (\text{background subtraction term})\,. \tag{5}$$

The Born arm enters the statistical error *only* through $C^{\rm rad}$ and $\varepsilon^{\rm Born}$, that is only in Form B and in the reported diagnostics. That is not a reason to generate less Born: see Sec. 5.4.

5 Facts checked in the actual samples, 6 November 2026

5.1 The branching ratio is not in the generator

```
grep -rniE "branching|0.0597|5.97" --include=*.py exclurad_py/   ->   nothing
```

`report.sigma_nb` is the **full J/ψ production cross section**. Every event is decayed to e^+e^- with probability one (`idecay = 2`). The data yield is an e^+e^- yield. Therefore $\mathcal{B}$ multiplies the Monte Carlo, once. Omitting it makes the answer wrong by 16.7.

5.2 The twins differ in two card fields, not one

From `prod_jpsi_f18in_20261009/{born,rad}/prov/*_043.card.txt`:

field	meaning	born	rad
4	`iborn`	1	0
18	`itarg`	0	1

This is *correct*: Born means no radiation of any kind, internal or external, so C^{rad} built from these twins contains target straggling *and* vertex radiation, which is what corrects data to Born. However the card header claims "IDENTICAL except iborn (field 4)", which is false; anyone diffing the two cards on that promise will conclude the sample is broken. The header should be fixed.

The `itarg = 0` point-vertex bug is **not** present in this sample: the fix `bbe6628` (2026-10-22 11:25) is an ancestor of the sample's recorded commit `8ac2fc0` (2026-10-22 12:36), so the Born arm has the proper 5 cm vertex spread rather than $v_z = 0$.

5.3 C^{rad} cannot be taken from `sigma_nb`

arm	`report.sigma_nb`
born	3.494470×10^{-4}
rad	3.368744×10^{-4}
ratio rad/born	0.9640

`sigma_nb` counts events *before* the acceptance preset, so any rad/Born ratio built from it is about 1.33 % high, and in any case it is a box average rather than a bin correction. C_i^{rad} must come from reconstructed yields inside the bin.

5.4 The arms are not yet at equal statistics

	born	rad
`out_inb`	1120	1167
`out_outb`	852	1180

The Born outbending arm is 28 % short, which caps the precision of both C^{rad} and $\varepsilon^{\mathrm{Born}}$. It should be finished before a systematic is quoted on either.

5.5 `iacc = jpsi` is a hard generation cut, not a flag

`events.py` executes `continue` on failure: the event is never written. The windows travel with the sample in the `.lund.info` sidecar:

particle	$\lvert p\rvert$ (GeV)	θ (deg)
proton	0.2–4.5	3–45
daughter	1.0–9.0	5–40
electron	0.1–2.5	2–20

If any analysis cut is *looser* than the corresponding preset edge, the Monte Carlo has exactly zero events there and the acceptance is wrong, not merely imprecise. Measured in a 2000-event sample, the live edges are the proton θ floor (minimum 3.013°, zero events below) and the daughter θ ceiling (maximum 39.96°). The proton $|p|$ window never bites: the kinematic minimum is 0.7156 GeV/c against a 0.2 GeV/c floor.

5.6 t from the proton, and the choice of t or t'

The proton t estimator has RMS 0.047 GeV^2 against 0.230 for the J/ψ one, a factor 5–6 better. The existing analysis ntuple carries `t_prime`, not `t`. Since $|t|_{\mathrm{min}}$ runs from 2.10 GeV^2 at threshold to 0.4536 at the top of the W range, bins in t' and in t map onto appreciably different regions. Fix the choice once and use it in the data, in both Monte Carlo arms, and in the model bin-averaging.

For reference, the recoil proton follows from t alone through $t = (p - p')^2 = 2M^2 - 2ME_p$, so

$$E_p = M + \frac{|t|}{2M}, \qquad |\mathbf{p}_p| = \sqrt{|t|}\,\sqrt{1 + \frac{|t|}{4M^2}}. \tag{6}$$

At the kinematic ceiling $W^2 = M^2 + 2ME_{\text{beam}} = 20.7717\ \text{GeV}^2$ ($W = 4.5576$ GeV),

$$|t|_{\min} = 0.4536\ \text{GeV}^2, \qquad |\mathbf{p}_p| = 0.7156\ \text{GeV}/c, \qquad \theta_p = 0.67^\circ, \tag{7}$$

so at the forward edge the recoil proton travels down the beamline and is undetectable; the 3° preset floor removes it. Over the whole box the recoil proton never exceeds $\theta_p = 25.3^\circ$: it is a Forward Detector particle everywhere, never a Central Detector one.

6 Systematics

Source	How to estimate
model shape inside the bin	spread over iterations; vary the Q^2 slope within its fit error
exclusivity cut	vary the MX^2 window; C^{rad} is the quantity that moves
signal extraction	window width; sideband versus fitted background
MC statistics	n^{rad} and n^{born} separately
efficiency scale factors	their own uncertainties, bin dependent
luminosity	from the data side
branching ratio	0.54 % relative
bin centring	centroid versus bin-averaged model

Table 2: Systematic uncertainties to carry through Eq. (3).

7 Ntuple branches the chain needs

The reader in `production/analysis/campaign_plots.py` uses

```
E_photon, Q2, W, t_prime, invariantMass,
scattered_theta, scattered_p, proton_theta, proton_p,
positron_theta, positron_p, electron_theta, electron_p,
time_ee, x_FT, y_FT
```

The data and both Monte Carlo arms must expose the same set, filled by the same code, or step 1 cannot be honoured.

Companion notes

- `docs/RC_METHOD_FOR_JPSI.md` — how the π^0 RC-method study was done and what to change for J/ψ.
- `docs/RC_METHOD_COMPARISON.md` — MC ratio versus the analytic `rc_factor`, and what the difference costs.
- `docs/rc_method_jpsi/rc_method_jpsi_report.tex` — the measured C^{rad} for this campaign, both methods.
- `docs/BORN_XSEC_EXTRACTION_JPSI.md` — the Markdown form of this note.